NGO XUAN CHIEN

Backend Software Engineer
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SUMMARY

Backend Software Engineer with 1.5+ years building core banking deposit products at HDBank on Thought Machine Vault Core. Works daily with Java, Spring Boot, Kafka, PostgreSQL and Redis, including End-of-Day batch jobs for 100,000+ deposit accounts and the go-live migration from the legacy core. Also builds microservices in Go with gRPC, event-driven design and observability.

WORK EXPERIENCE

Software Engineer - HDBank *Feb 2025 - Present*

Project: Core Banking Modernization (Thought Machine Vault Core) - Deposit team of 6

- Develop term deposit (savings) products with the Deposit team, covering the full account lifecycle: opening, top-ups, partial and early withdrawal, maturity settlement and rollover, holds, joint holders and holiday maturity, using Java, Spring Boot, PostgreSQL, Redis and Kafka.
- Own the automatic savings product end to end as its sole developer: scheduled debits from a customer's current account, each opening a new term deposit, through to maturity and rollover.
- Develop and maintain End-of-Day (EOD) batch jobs that accrue interest and settle maturities for 100,000+ deposit accounts daily, where every calculation must reconcile exactly.
- Took part in the go-live migration from the legacy Symbols core banking system, configuring interest rate tables, savings products and automatic savings accounts.
- Own interest calculation in production: periodic, upfront and at-maturity payouts, tiered and promotional rates, early-withdrawal interest, expense accrual, and day-count and rounding rules.

Trainee (Learner to Master Program) - HDBank *Aug 2024 - Nov 2024*

- Built a Notification Service in Spring Boot (Java 21, virtual threads, hexagonal architecture, Spring Modulith) that consumes core banking error events from Kafka, routes them by type (transfers, postings, settlements), stores them in PostgreSQL with Redis caching, and alerts via Email, Telegram and Kafka topics.
- Instrumented the service with OpenTelemetry, shipping logs, metrics and traces to Prometheus, Grafana, Loki, Tempo, Jaeger and Zipkin; worked hands-on with Kubernetes, Docker and Kafka Streams.

Developer Intern - DEK Technologies *Jun 2023 - Aug 2023*

- Developed an Android application for voice calls, video calls and messaging over the IP Multimedia Subsystem (IMS).
- Set up Jenkins to run automated tests for the application.

PROJECTS

Multi-vendor Marketplace - Go, gRPC, Kafka, PostgreSQL, React *Sep 2026 - Present*

- Design and build a Shopee-style marketplace of 6 Go microservices (identity, catalog, order, payment, ledger, gateway): REST at the edge, gRPC between services, one database per service.
- Implement multi-shop checkout that splits a cart into one order per shop, idempotent and oversell-safe under concurrent requests; events are published to Kafka via a transactional outbox.
- Integrate VNPay/Stripe sandbox payments with escrow, a double-entry ledger, seller wallets and payouts; secured with RS256 JWT, traced with OpenTelemetry and Jaeger, and tested in CI with Testcontainers and Playwright.

- Handle payment failure paths: a reconciler recovers dropped webhooks, and a refund worker reverses cancelled and duplicate payments.

ZingMp3 Backend - Java, Spring Boot, Kafka, Keycloak, Docker *Apr 2024 - May 2024*

- Built a RESTful API for music, albums and playlists as microservices behind an API Gateway with Eureka service discovery.
- Applied CQRS and Event Sourcing with Apache Kafka, splitting each domain service into a core module and an event-handler module.
- Secured APIs with Keycloak; containerized the stack with Docker Compose and automated builds with a Jenkins pipeline.

TECHNICAL SKILLS

Languages: Java, Go, C++, Python, JavaScript/TypeScript
Frameworks: Spring Boot, Spring Cloud (Gateway, Eureka), Keycloak, ReactJS
Data & Messaging: PostgreSQL, Redis, Kafka, Kafka Streams
DevOps & Observability: Docker, Kubernetes, Jenkins, Git, Maven, Linux, OpenTelemetry, Prometheus, Grafana, Loki, Tempo, Jaeger, Zipkin
Banking Domain: Core Banking, Thought Machine Vault Core, Term Deposits, Interest Calculation, EOD Batch Processing
Concepts: Microservices, Spring Modulith, Hexagonal Architecture, Event-Driven Architecture, CQRS, Event Sourcing, Transactional Outbox, REST, gRPC
Testing: Unit Testing, Integration Testing (Testcontainers), E2E Testing (Playwright)

EDUCATION

Ho Chi Minh University of Science - HCMUS *2020 - 2024*
Bachelor of Science in Information Technology GPA: 8.0/10
English: VSTEP B1 (CEFR B1)

ACHIEVEMENTS

Consolation Prize, Vietnam National Olympiad in Informatics (VOI), school year 2018 - 2019.